SUN SENSOR – CHARACTERIZATION AND TESTING

4.1 Introduction and Objective

In satellite attitude determination and control systems (ADCS), the sun sensor plays a vital role in accurately determining the sun vector relative to the satellite body frame. This information is crucial for stabilizing and orienting the spacecraft, particularly during sun-pointing operations and solar array alignment. The Position Sensitive Detector (PSD)-based sun sensor provides an analog output proportional to the position of the incident light spot on its active surface, enabling precise angular measurement with high resolution and low noise.

The primary objectivee of this work is to design, interface, and experimentally characterize a PSD-based analog sun sensor similar in architecture to the FS1 model. The aim is to understand its working principle, electrical behavior, and angular response characteristics under controlled laboratory conditions. The developed sensor will later be used for CubeSat-scale ADCS applications or deployable optical telescope systems requiring fine sun-tracking capability.

4.2 Working Principle of PSD-Based Sun Sensor

A Position Sensitive Detector (PSD) registers the position of an incident light spot on a continuous active surface. When a collimated (or non-collimated) beam falls through an aperture onto the PSD, the light creates a small spot whose location depends on the incident direction of the beam. By measuring the photocurrents (or voltages) at the four PSD electrodes (Vx1, Vx2, Vy1, Vy2), the spot centroid can be inferred and mapped to a sun-vector direction.

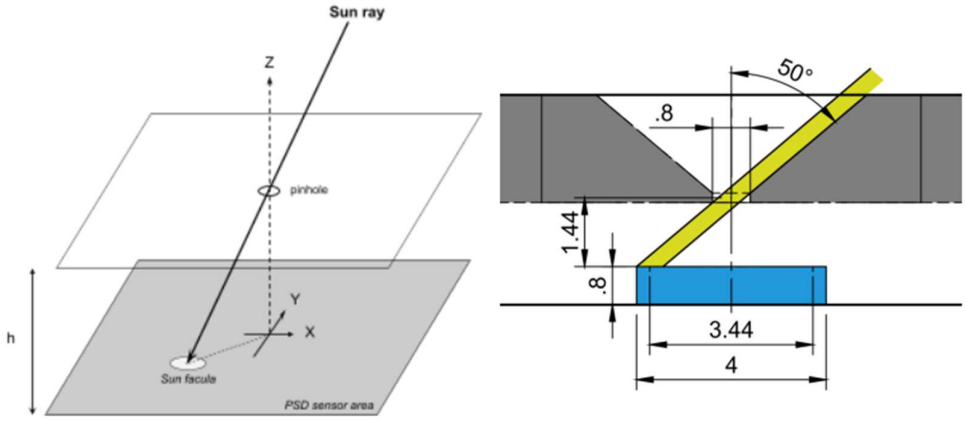


Fig. 4.1 (replicated as standalone) PSD Sun Sensor Working Principle

4.2.1 Onboard Computation (MSP430FR2311)

All primary signal conditioning and geometric computations are performed on the embedded MCU (MSP430FR2311). The firmware acquires the four corner voltages and computes normalized X and Y coordinates:

$$X = (Vx2 + Vy1 - Vx1 - Vy2) / Sum$$
$$Y = (Vx2 + Vy2 - Vx1 - Vy1) / Sum$$
$$\textit{where } Sum = Vx1 + Vx2 + Vy1 + Vy2$$

These normalized coordinates are then converted to incident angles using calibrated polynomial transformations. Temperature compensation and health/status flags are also generated onboard and packed into I²C-readable registers.

4.2.2 From Spot Position to Sun Vector

The geometric mapping from PSD coordinates to the sun vector uses the effective focal length (aperture-to-PSD distance, f). For small incident angles, the mapping is approximated as:

$$\theta x = \arctan(X / f) \ , \ \theta y = \arctan(Y / f)$$

In practice, the MCU implements a calibrated polynomial function $\theta = f_calib(X)$ to compensate for non-linearities, wide field-of-view effects, and optical distortions introduced by the aperture and housing. Calibration coefficients are stored in firmware and applied in real-time.

4.3 Sensor Design and Configuration

4.3.1 Structural Overview

The sensor assembly follows the FS1 PSD-based architecture. It comprises an optical entrance aperture defining the effective focal length, a PSD active element (Hamamatsu S5990-01), front-end analog electronics (trans-impedance amplifiers), and an embedded MSP430FR2311 microcontroller for signal processing. The design emphasizes mechanical stability, light-tight housing, and thermal management for repeatable measurements.

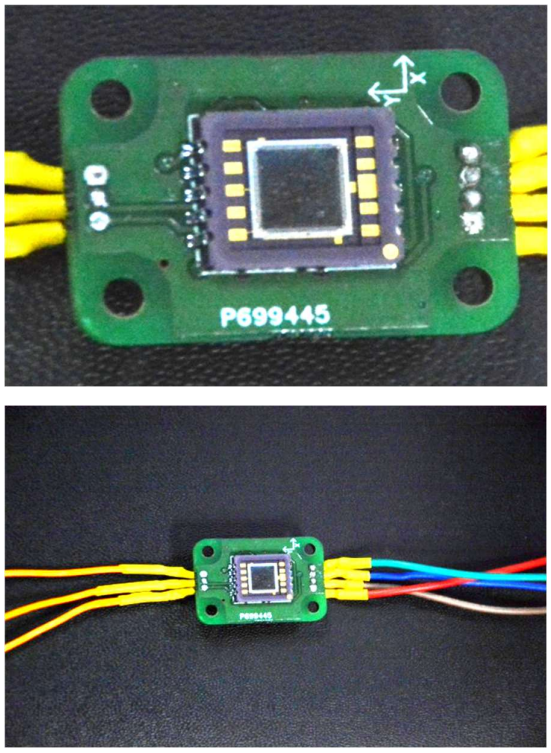


Fig. 4.1 PSD Sun Sensor Housing and Hamamatsu S5990-01 PSD Device

4.3.2 Electronics and Signal Conditioning

The analog front-end conditions the low-level photocurrents into measurable voltages for ADC sampling by the MCU. The main elements include: trans-impedance amplifiers (TIA) for each PSD electrode converting current to voltage; low-pass filtering to reduce high-frequency noise; offset correction and gain calibration circuits; and a temperature sensor near the PSD for compensation in firmware.

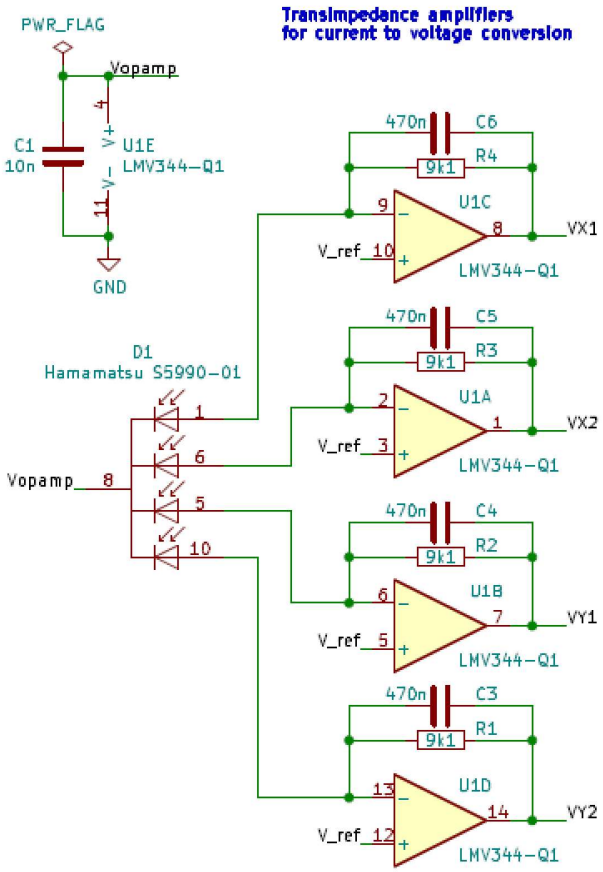


Fig. 4.2 Schematic of Trans-Impedance Amplifiers (TIA) for PSD Signal Conditioning

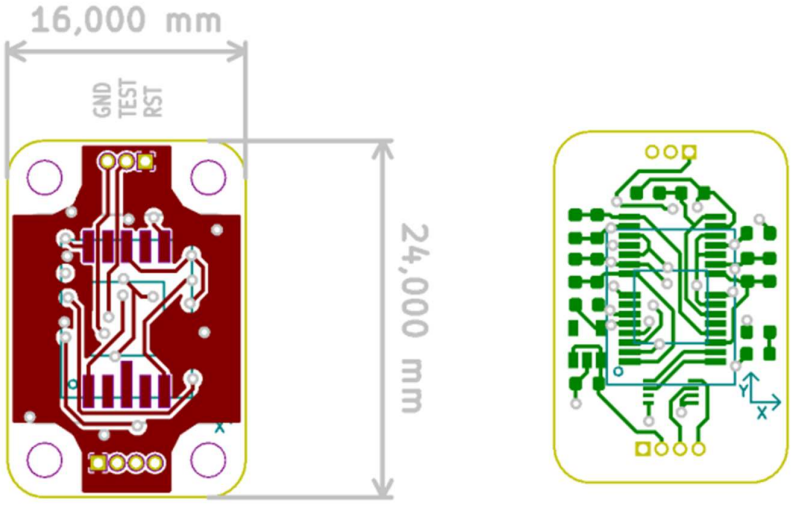


Fig. 4.3 PCB Layout: Front-Side and Back-Side of the Sun Sensor Hardware

4.3.3 Mechanical Housing and Mounting

The sensor housing is precision-machined to maintain exact aperture-to-PSD spacing and to provide a rigid, light-tight enclosure. Key features include: mounting holes compatible with the stepper motor fixture used for angular characterization; internal baffling with matte-black surfaces to minimize stray light reflections; and a thermal path for heat dissipation with a local temperature sensor near the PSD.

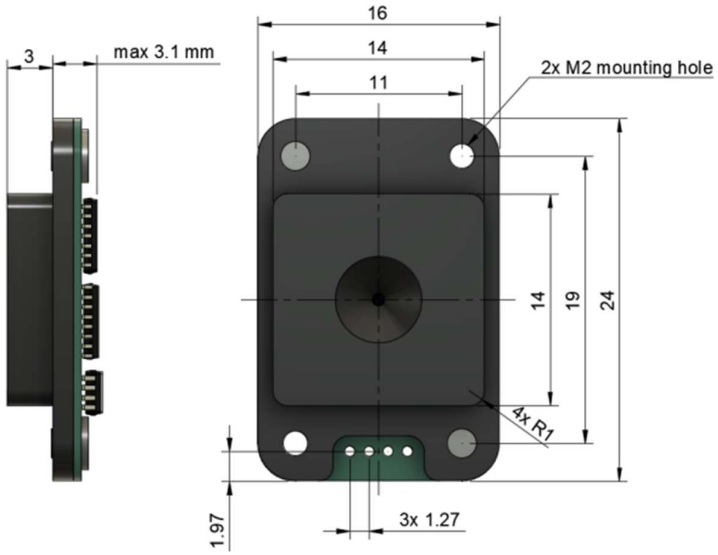


Fig. 4.4 Panel Cutout Dimensions for Sun Sensor Module. Preferred mounting: top-right and bottom-left holes.

4.4 Electrical Interface and Communication

The sensor communicates via I²C (3.3V logic) and is interfaced with a Raspberry Pi 4 acting as the I²C master. Pull-up resistors (4.7 kΩ) are required on both SDA and SCL lines. The sensor exposes a register map that includes: RAW_X, RAW_Y (raw PSD electrode voltages/ADC counts); V_SUM (sum reference voltage); ANGLE_X / ANGLE_Y (computed incident angles in degrees); TEMP (sensor temperature); and STATUS/CONFIG registers.

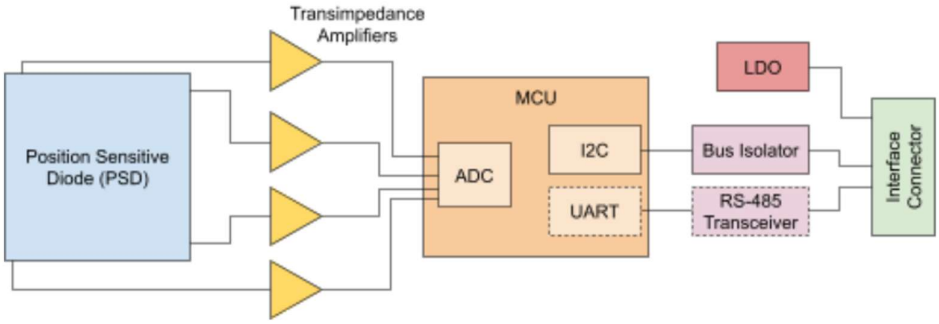


Fig. 4.5 Block Diagram of Sensor Electrical Components and Connections

4.4.1 I²C Connection (Raspberry Pi 4)

The hardware connections for the Raspberry Pi 4 interface are:

- SDA → Physical Pin 3 (GPIO2, SDA1)
- SCL → Physical Pin 5 (GPIO3, SCL1)
- Power: 3.3V or 5V as specified in sensor design document
- GND → Any ground pin

The I²C bus speed is set at 100 kHz for reliable communication. The device address is confirmed using the 'i2cdetect -y 1' command on the Pi. All register reads are performed as multi-byte transactions to ensure data consistency between channels.

4.5 Static Test: Experimental Setup and Observations

4.5.1 Setup Description

The static test evaluates the sensor's output stability and noise characteristics when pointed at a fixed light source. The sensor was mounted on a rigid stand with no mechanical play. A non-collimated LED/lamp source was placed at varying distances and alignments relative to the sensor normal. The ambient light was reduced to minimize stray illumination effects.

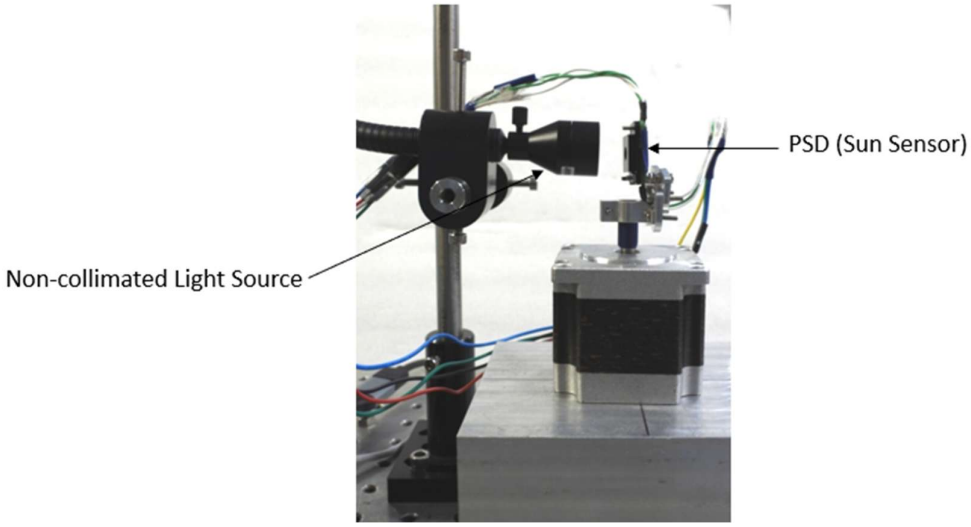


Fig. 4.6 Static Test Setup – Sun Sensor Fixed on Stand Facing the Light Source

4.5.2 Key Observations

The static experiments revealed the following key insights:

- Distance dependence: When the light source was placed far from the sensor, the angle and voltage readings showed large fluctuations and occasional abrupt jumps, indicating high sensitivity to the divergence angle of the non-collimated source.
- Close-range stability: When the source was positioned close to the sensor and roughly on-axis, the output became significantly more stable with small, manageable fluctuations.
- Averaging improvement: Averaging 40–50 consecutive samples reduced angle fluctuations to less than 1°, making the reading practically useful for ADCS test bench applications.
- Intensity sensitivity: Reducing light intensity increased noise and fluctuation magnitude, confirming the importance of adequate illumination for reliable sun vector estimation.

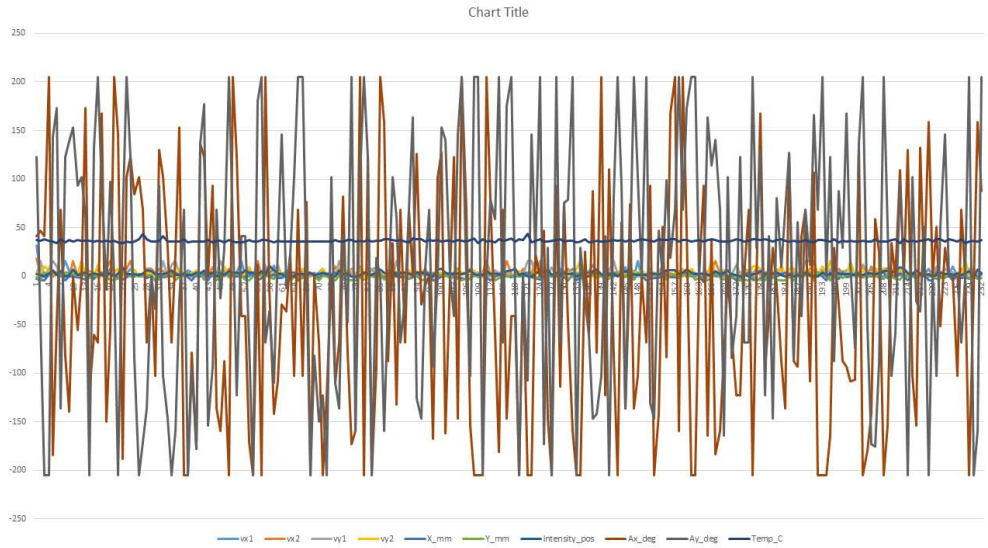


Fig. 4.7 Plot: Sensor Output When Non-Collimated Light Source is far – High Fluctuation Observable

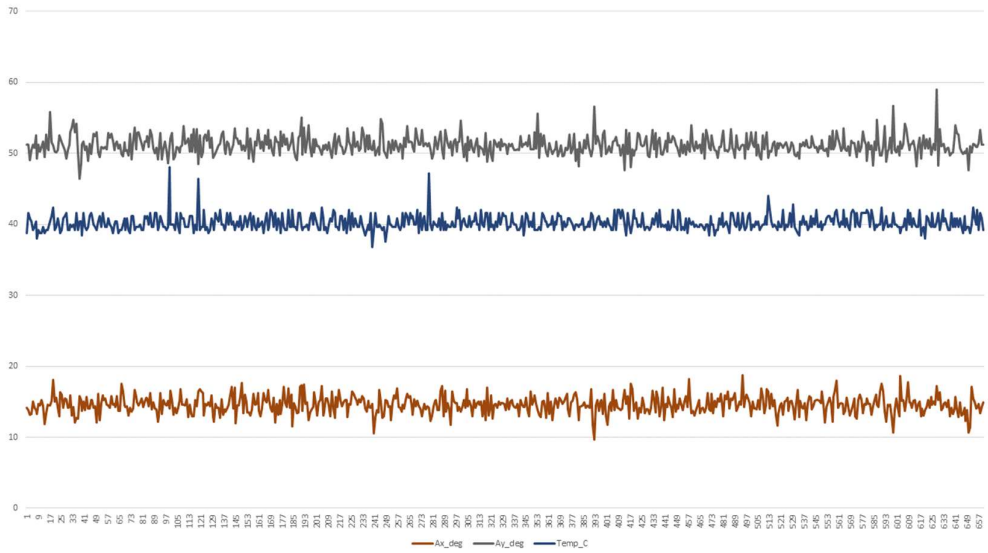


Fig. 4.8 Plot: Sensor Output When Light Source is Close – Improved Stability

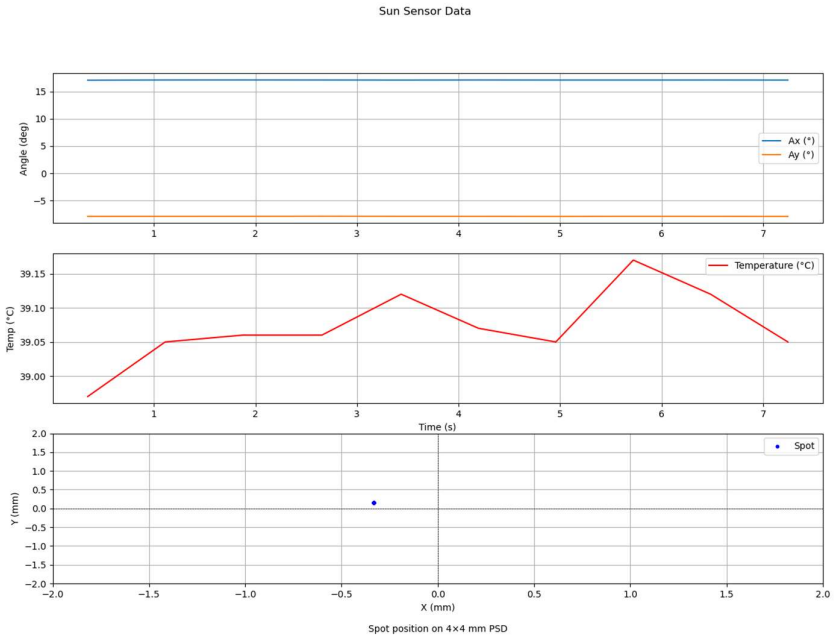


Fig. 4.9 Plot: Close Source with Averaged Readings (n=40–50) – Fluctuation <1°

4.6 Angular Characterization: Setup and Methodology

4.6.1 Objective

The angular characterization experiment determines how accurately the sensor responds to changes in the incident light angle, and verifies the consistency and linearity of readings across repeated measurement trials. This is essential for establishing a reliable calibration model.

4.6.2 Setup Description

The sensor was mounted on a stepper motor-driven rotation stage (Microstep Driver DM542), enabling precise angular increments about one axis. A non-collimated LED source remained fixed in space at a constant distance. The zero-angle reference was defined when the sensor face was perpendicular to the beam. An automated Python script controlled the stepper, acquired sensor readings, and logged all data.

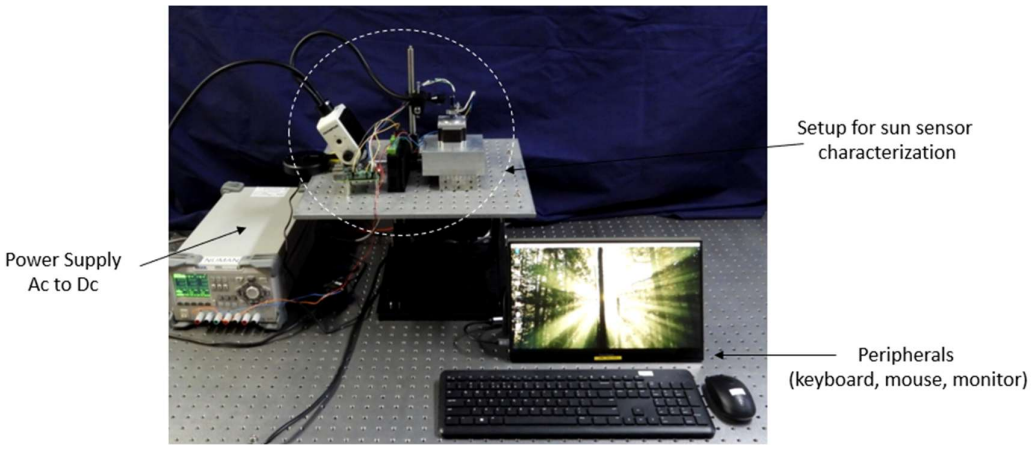


Fig. 4.10 Experimental Setup for Angular Characterization – Stepper-Mounted Sensor

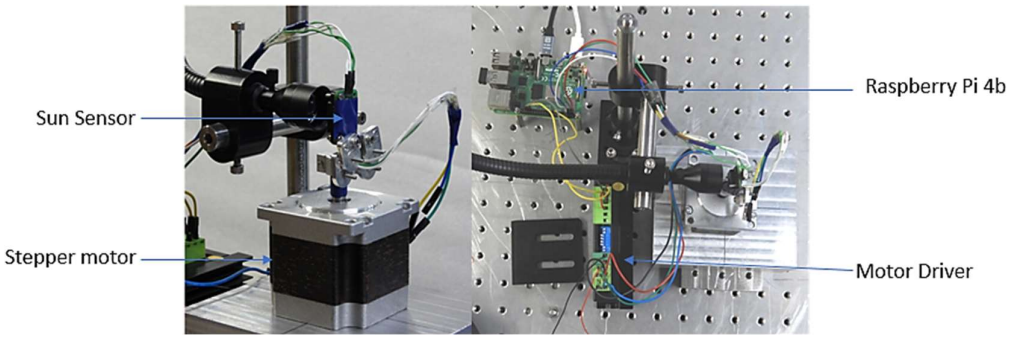


Fig. 4.11 Angular Characterization Setup

4.6.3 Procedure

The characterization procedure was as follows:

1. Initialize the stepper motor with the current sun sensor angle output as the reference.
2. Rotate the sensor in small increments of 0.225° (microstep resolution).
3. After each rotation, pause for ~0.1 s settling time and acquire 40–50 sensor readings.
4. Continue rotation up to ±50° from the zero reference angle.
5. Repeat the entire run 5 to 10 times to assess measurement repeatability across trials.

4.6.4 Results – X-Axis Rotation

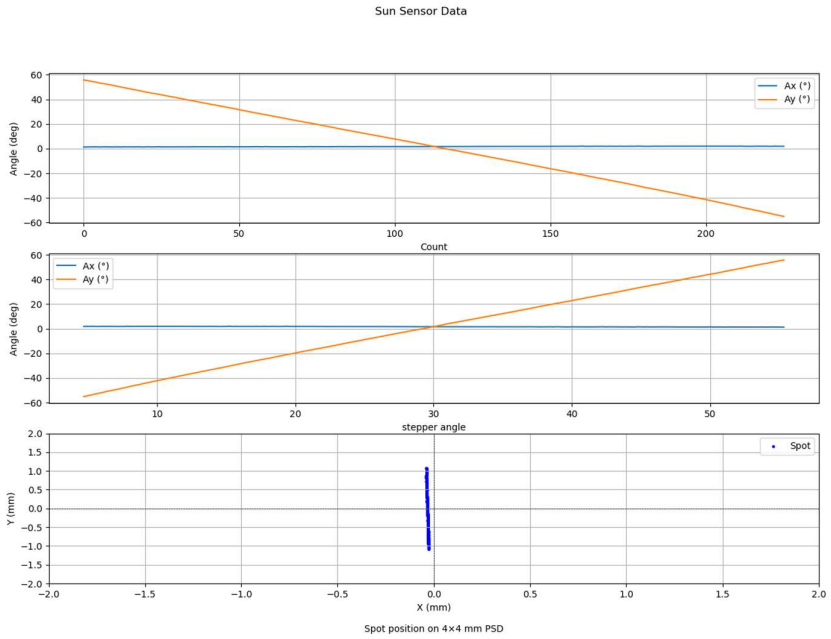


Fig. 4.12 Test 1: Sensor Angle vs. Actual Stepper Angle Change (X-Axis Rotation)

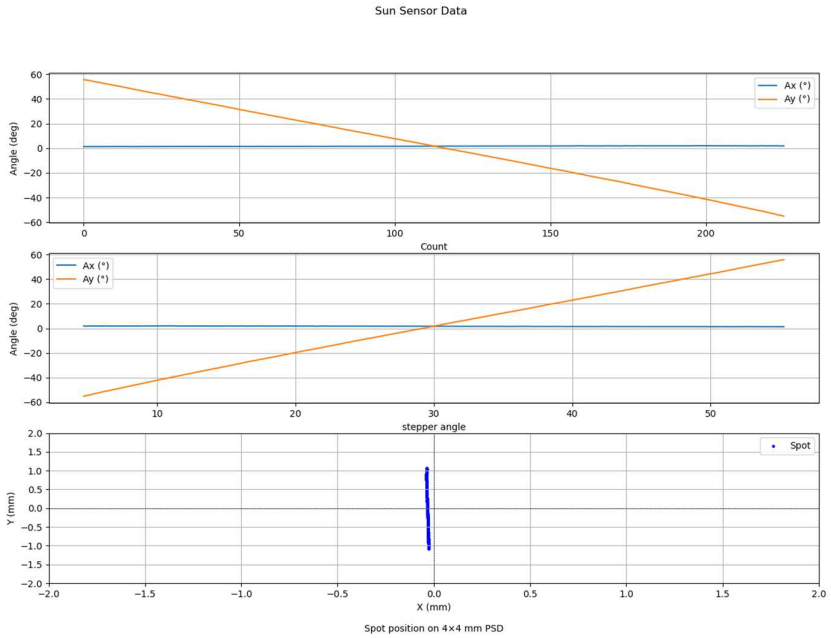


Fig. 4.13 Test 2: Sensor Angle vs. Actual Stepper Angle Change (X-Axis Rotation)

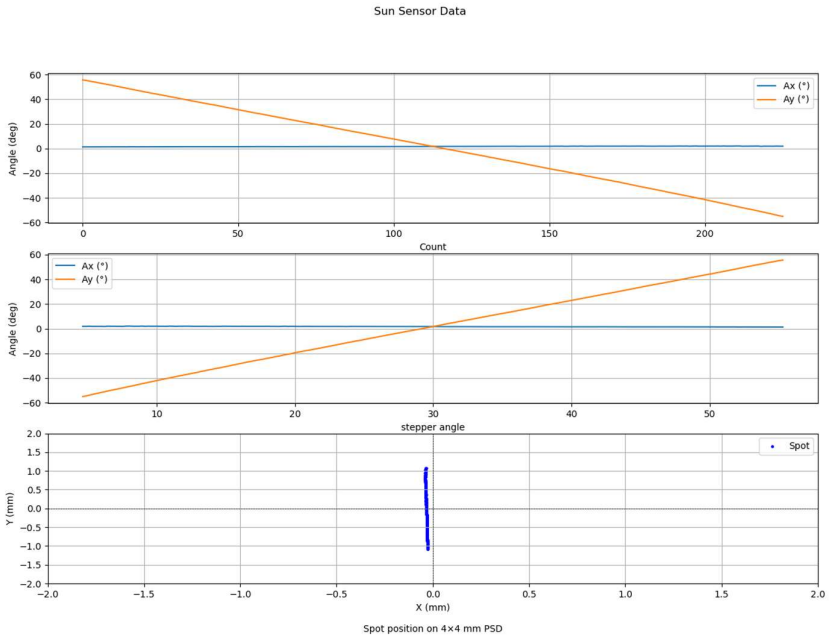


Fig. 4.14 Test 3: Sensor Angle vs. Actual Stepper Angle Change (X-Axis Rotation)

4.6.5 Results – Y-Axis Rotation

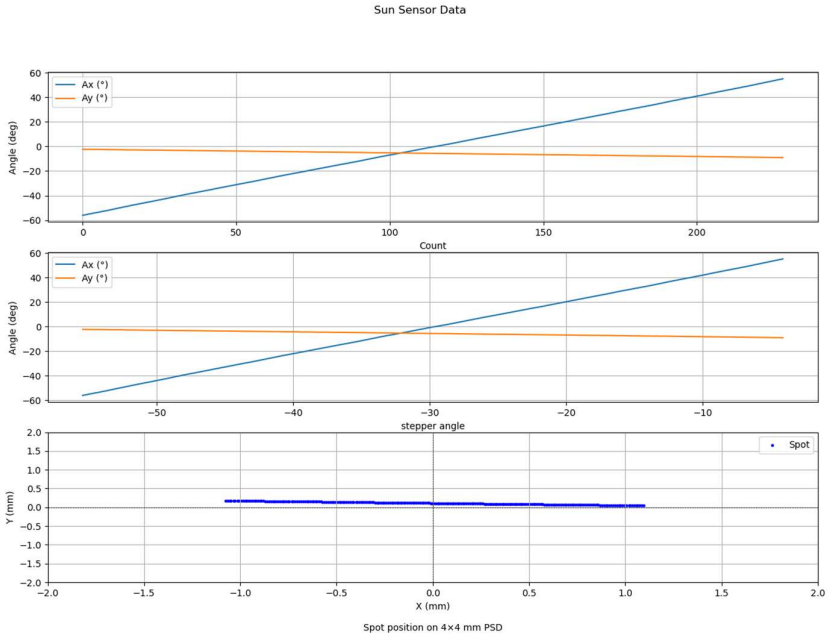


Fig. 4.15 Test 1: Sensor Angle vs. Actual Stepper Angle Change (Y-Axis Rotation)

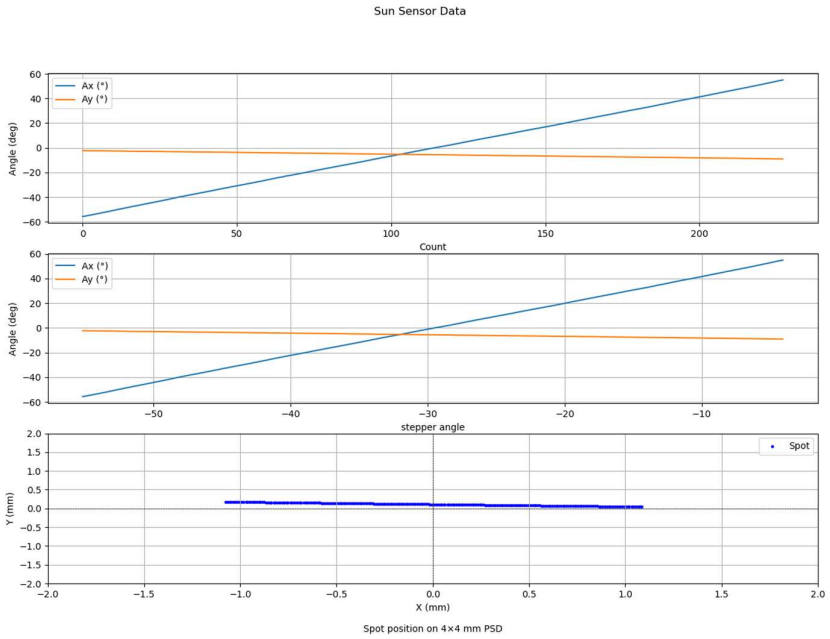


Fig. 4.16 Test 2: Sensor Angle vs. Actual Stepper Angle Change (Y-Axis Rotation)

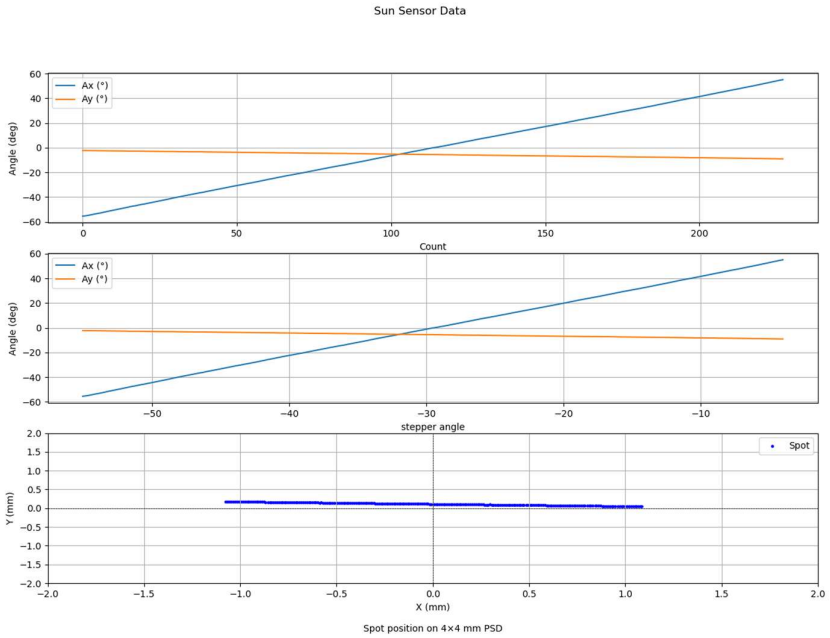


Fig. 4.17 Test 3: Sensor Angle vs. Actual Stepper Angle Change (Y-Axis Rotation)

4.7 Analysis and Discussion

4.7.1 Angular Response Behaviour

A broadly linear trend was observed between mechanical stepper angle and sensor-reported angle; however, the response slope exceeded unity. A ~50° mechanical

rotation corresponded to approximately 100° change in sensor-reported angle, indicating an optical magnification factor of approximately 2×. The primary cause is the use of a non-collimated light source: beam divergence causes the angular subtended at the PSD surface to change nonlinearly with sensor tilt, leading to an apparent magnification of angle changes. The response curve remained consistent across repeated trials, confirming good repeatability of the sensor electronics and signal chain.

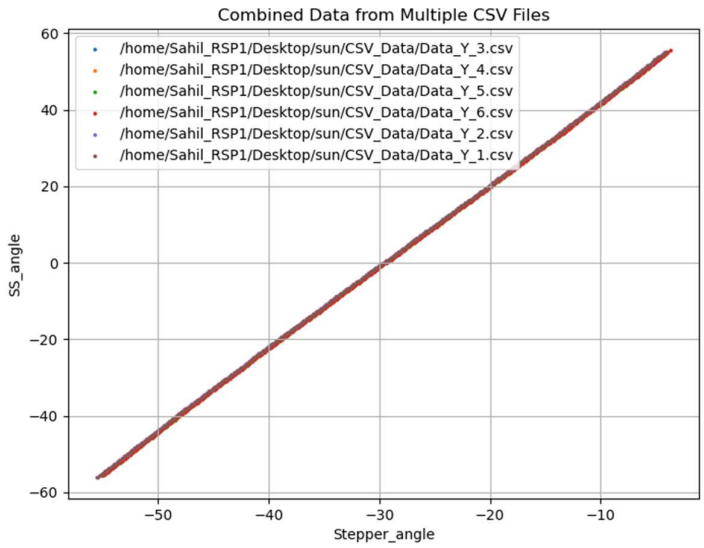


Fig. 4.18 Polynomial Curve Fitting – Sun Sensor Angle vs. Stepper Angle (Y-Axis)

4.7.2 Polynomial Fitting and RMSE Evaluation

Measured data from multiple trials was fitted with low-order polynomial models (2nd and 4th order) to capture the non-linear mapping between mechanical and sensor angles. The Root Mean Square Error (RMSE) was computed for each fit to quantitatively assess stability across trials.

Table 4.2 Polynomial Fit Equations and RMSE – Y-Axis (RY) Rotation

Trial	Fit Equation	RMSE (°)
Trial 1	$0.002007x + 2.282x^2 – 64.8$	0.2904
Trial 2	$0.002042x + 2.285x^2 – 64.63$	0.2927
Trial 3	$0.002002x + 2.281x^2 – 65.00$	0.2941

Table 4.3 Polynomial Fit Equations and RMSE – X-Axis (RX) Rotation

Trial	Fit Equation	RMSE (°)
Trial 1	$0.0002371x + 2.131x^2 + 63.01$	0.2466
Trial 2	$0.0002167x + 2.132x^2 + 62.79$	0.2474
Trial 3	$0.0002326x + 2.131x^2 + 62.79$	0.2523

4.7.3 Repeatability and Stability

For each repeated test run, the same mechanical orientation produced nearly identical sensor readings with variations consistently below 1–2°. Temperature drift had negligible impact over the short test duration (less than 0.2°C variation). This demonstrates excellent consistency of the internal circuitry and PSD response. The characteristic response curve, despite its non-unity slope due to non-collimated source geometry, can be used to calibrate true incident angles through polynomial mapping.

4.7.4 Recommendations

- Use a collimated light source for absolute angular calibration to remove geometric/parallax errors and verify linearity.
- Maintain light source close to the sensor and well-aligned for low-noise static readings.
- Continue using n = 40–50 sample averaging or implement a lightweight moving-average filter for real-time applications.
- Log raw samples and temperature for each test session to separately quantify thermal drift effects.